

Optical Hotspot for Adaptive Pencil-/Wide-Beam Formation for Fiber-Wireless Connectivity with Integrated User Localization

F. Honz, and B. Schrenk

AIT Austrian Institute of Technology, Giefinggasse 4, 1210 Vienna, Austria, florian.honz@ait.ac.at

Abstract We propose an optical hotspot for adaptive pencil-/wide-beam formation, enabling user localization and the establishment of an optical fiber-wireless connection by means of tunable optics. We confirm stable optical wireless operation over a free-space link reach of 107 m. ©2026 The Author(s)

Introduction

Optical wireless communication (OWC) enables the extension of high-capacity links over the air [1]. It operates on a license-free basis and can build on a rich optical spectrum while enhancing robustness to electro-magnetic interference. The properties of light allow the adoption of dedicated and private OWC links to individual users, provided that the involved pencil beams can be established and maintained as direct line-of-sight connection under low overhead concerning the complexity of the transmitting and receiving OWC sub-systems. The development of optical beamforming technologies [2-7] tackles this challenge through demonstration of large optical antenna arrays that can be hosted on a few square-millimeters [2, 3, 5-7]. At the same time, means for localizing communication end-points become indispensable when serving mobile users. Optical retro-reflectors [8-10], dedicated beacon emitters [11] or direct beam sensing [12] have been employed to ensure optimal over-the-air coupling between OWC transceivers.

This work shows simultaneous beamsteering and field-of-illumination tuning for adaptive pencil-/wide-beam generation by an optical focal plane array (FPA) hotspot with tunable optics. We evaluate the hotspot in a fiber-wireless OWC link and confirm successful user localization and stable TCP transmission at 605 Mb/s over a 107-m out-door reach.

Optical Hotspot for OWC and MU Localization

We employ an FPA beamformer at the optical hotspot (OHS) that represents the demarcation point between the fiber infrastructure and the free-space segment (Fig. 1). It builds on a 91-element photonic lantern that acts as interposer between an arrayed waveguide grating (AWG) and the focal plane of an optical lens. Its single-mode cores represent the antenna elements of the OHS, which are hexagonally arranged at the focal plane with a spacing of $35.1 \mu\text{m}$ to enable wavelength-tuned beamsteering in combination with the 50-GHz spaced AWG beamforming network [13]. The 10-mm lens (TL) of the OHS is laid out as focus-tunable element (Optotune EL-10-30), whose focal power (FP) range between

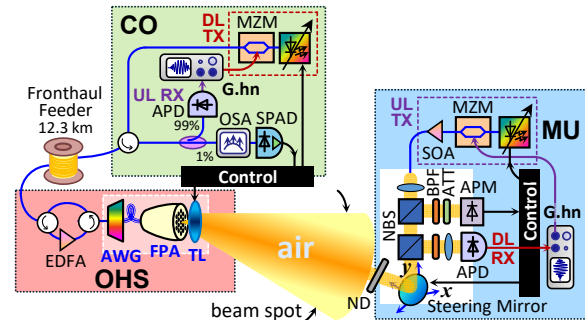


Fig. 1: Fi-Wi setting with OHS integrating beam steering and field-of-illumination adaptation for OWC and user localization.

10 to 24 dpt (corresponding to focal lengths from 100 to 41.7 mm) allows an adaptation of the beam spot size and thus the beam overlap and field-of-illumination. Due to the distance of 43.2 mm between the focal plane across the photonic lantern and the lens, a FP of 23.15 dpt results in a well-collimated beam. The localization of mobile units (MU) within the field-of-view of the OHS is achieved through highly-sensitive signal capture across the FPA plane. The required spectral acquisition is performed by a remoted low-light spectrum analyzer including a single-photon avalanche detector (SPAD), both of which are hosted at the central office (CO) that is connected to the OHS through a 12.3 km fronthaul fiber. The excellent sensitivity of the SPAD captures the FPA-specific signatures resulting from amplified spontaneous emission (ASE) noise of a free-running semiconductor optical amplifier (SOA) at the MU, which acts as beacon for its localization through the OHS. For the MU, whose simplification is not a prime focus of this work, we employed a tunable MEMS mirror (Mirrorcle A7B2.1, mirror diameter of 3.6 mm) to obtain 2D beamsteering functionality. Though not representing a solid-state solution, tunable mirrors have proven fast response times in real-time OWC applications [14]. Placed at the air interface of the MU, it connects the OWC channel to a transmitter / receiver pair, whereas the latter includes a free-space GHz-grade avalanche photodetector (APD) for data reception. The APD receiver shares its optical axis with a co-linear, highly-sensitive avalanche photodiode monitor (APM) with a low NEP of $7.5 \text{ fW}/\sqrt{\text{Hz}}$ to locate the

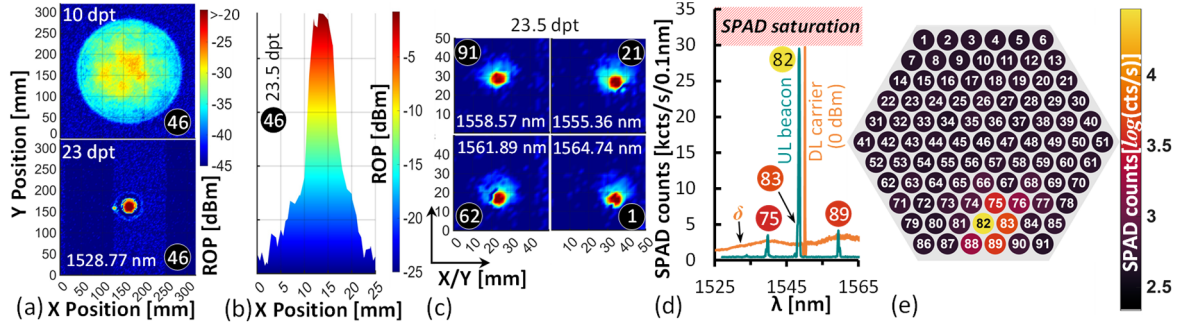


Fig. 2: (a) FP-dependent optical power distribution across the OWC beam and (b) beam profile for FP = 23.5 dpt. (c) Beam steering through the OHS. (d) Low-light spectroscopy of UL beacon received by the OHS and (e) mapping over its focal plane.

ASE beacon emitted by the OHS. 50-nm wide optical C-band bandpass filters (BPF) centric to 1550 nm were included to avoid saturation of APD and APM due to solar irradiance.

The MU and OHS perform localization by first aligning the MEMS mirror of the MU towards the OHS with the help of the APM. The MU then emits its own beacon, enabling the OHS to subsequently infer on the position of the MU and the optical antenna element that serves the user.

The OWC channel is first evaluated at lab-scale before migrating the setting to an out-door scenario. We employed tunable sources together with Mach-Zehnder modulators (MZM) to imprint OOK data at a variable data rate from 0.1 to 1 Gb/s to probe OWC connectivity over the optical link budget. To allow for finer granularity in rate-adaptive OWC settings, we additionally utilized G.hn as a 200-MHz wide multi-carrier solution. End-to-end link stability is further evaluated through real-time TPC data transmission over Ethernet. Since no tunable L-band source was available, the evaluation was conducted in half-duplex mode – even though we could in principle take advantage of the cyclic AWG nature for the purpose of full-duplex operation [13]. The launch levels at the OHS and MU were fixed at 7 and 10 dBm respectively, in compliance with eye-safety requirements.

Characterisation of Wide-/Pencil-Beam OWC

First, we characterized the achievable beam spot sizes through lens tuning at the OHS. For this, we used an in-door setup with a distance of 2.4 m between lens and APM at the MU, with the latter being mounted on a 2D beam profiler. A single beam was emitted by the OHS via its central

antenna element (#46) at 1528.77 nm. The measured optical power distribution for FPs of 10 and 24 dpt is shown in Fig. 2a. As can be seen, the total illuminated area where the APM acquires a signal above its noise floor features a diameter of ~247 mm for a highly diverging beam at FP = 10 dpt. For a FP of 23 dpt the diameter of the illuminated area is ~38 mm. Since highly collimated beams lead to a saturation of the APM and thus a flat-top beam profile at the center while still capturing the faint onset of illumination as a result of the excellent sensitivity of the APM, the spot in Fig. 2a is artificially broadened. Comparison with a slightly changed FP of 23.5 dpt reveals the Gaussian beam profile and actual FWHM beam diameter of 4.5 mm (Fig. 2b). In summary, the tunable lens offers a 6.5-fold range extension for the OHS' field-of-illumination through focal-length tuning.

Figure 2c shows the wavelength-tuned beamsteering capabilities of the FPA for four exemplary antenna elements on the outer "ring" of its cross-section. For the tested FP of 23.5 dpt the covered field-of-illumination is ~18 mm.

We have then evaluated the impact of the beam spot configuration on data transmission. After the APM-assisted alignment of the MU's MEMS mirror for which the beacon is sourced by the noise emission of the OHS' booster EDFA, the broadband ASE emission of the MU's booster SOA was employed as beacon to determine the optimal FPA element at the OHS through the SPAD-assisted low-light spectroscopy at the CO (Fig. 2c). The detection efficiency of this measurement was artificially reduced to -37 dB to avoid saturation of the SPAD by broadband

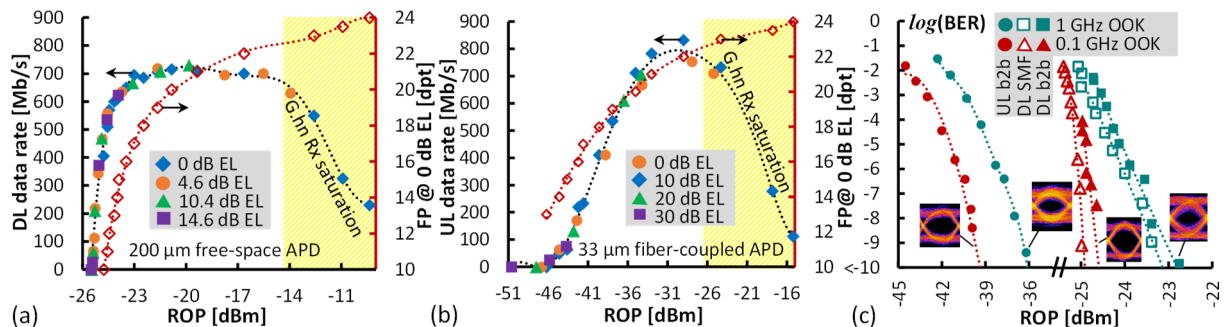


Fig. 3: (a) DL and (b) UL G.hn rates over lab-scale OWC link, and (c) BER performance for 0.1 and 1 Gb/s OOK modulated UL / DL.

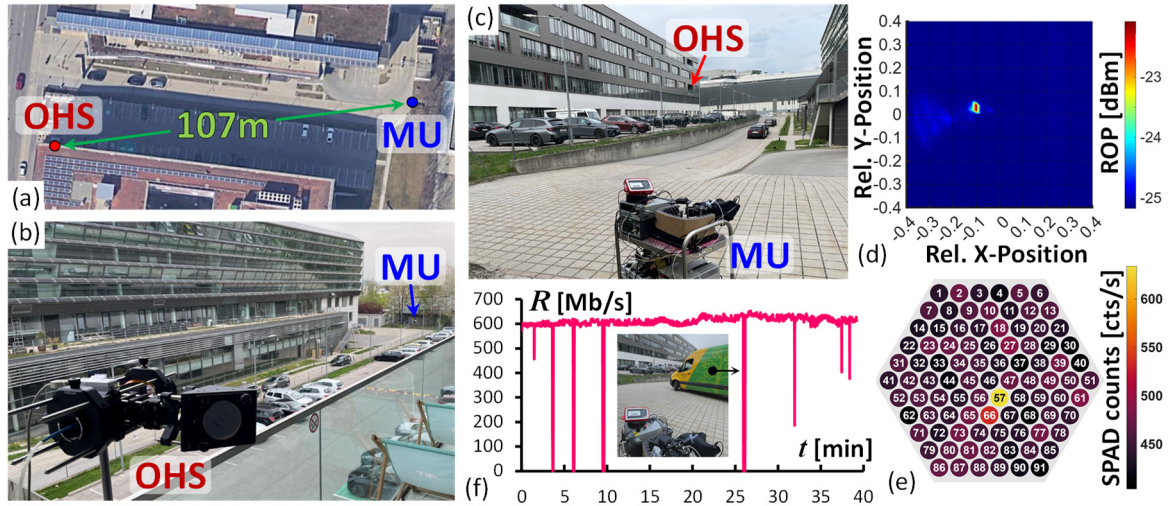


Fig. 4: (a) Out-door OWC link between OHS/CO and MU. (b) OHS installation overlooking the parking space. (c) View of OHS from site of MU. (d) OHS localization at the MU and (e) MU localization at the OHS. (f) Downlink TCP data throughput.

background noise, created by Rayleigh/Raman scattering and solar photons. A clear peak at 1548.5 nm due to the UL beacon determines the optimal FPA antenna element (#82) that links the MU to the OHS (Fig. 2d). The remarkable sensitivity of the SPAD-based monitor allows to identify further antenna elements (#83, 89, 75) to which the MU beacon is partially coupled, potentially enabling center-of-gravity schemes to further optimize beam tracking. Once a DL signal with a launch of 0 dBm is activated, noise is generated due to Raman scattering along the feeder fiber, elevating the SPAD background to ~ 2.5 kcts/s. However, the SNR of the beacon remains as high as 10.7 dB.

Figures 3a and 3b report the down- and uplink G.hn line rates as a function of the FP through the corresponding received optical power (ROP) level. Despite the small mirror size of 3.6 mm that limits the receiver aperture at the MU, DL transmission can be maintained at >500 Mb/s for ROPs larger than -24.6 dBm, corresponding to a FP of ~ 11 dpt or a wide beam diameter of ~ 235 mm. This point is taken as a reference for an excess path loss (EL) of 0 dB. We then increased the EL through insertion of neutral density (ND) filters in front of the MU aperture, which allows us to probe the compatible optical budget when using narrower pencil beams. For an EL of 14.6 dB a FP of 23.5 or a beam diameter of 4.5 mm was required to retain data rates of >500 Mb/s.

For the UL scenario, data transmission beyond 500 Mb/s can be established for ROPs above -38.1 dBm due to the higher sensitivity of the employed fiber-coupled APD receiver. Since the UL beam diameter of 3 mm was optimized to fit the mirror size at the MU, the geometry necessitates higher FPs of >18.8 dpt to retain efficient fiber coupling through the FPA beamformer at the OHS. For both, UL and DL, a reduction of the achievable data rate at high optical power levels is noticeable due to

saturation of the fixed-gain receiver setting.

Simpler OOK data transmission at 0.1 and 1 Gb/s requires reception sensitivities of -24.6 (-24.9) and -23 (-23.3) dBm in a back-to-back (fiber) scenario (Fig. 3c). The UL sensitivity for a BER of 10^{-9} is -39.7 / -36.4 dBm for 0.1 / 1 Gb/s OOK transmission.

Out-Door OWC Link and MU Localization

We then migrated the setup to an out-door environment (Fig. 4a) to evaluate DL connectivity over an extended OWC reach. For this, the OHS was placed at an elevated position (Fig. 4b) overlooking a parking space at which the MU was located (Fig. 4c) at a distance of more than 100 m from the OHS. We performed localization at the MU and OHS, yielding the beacons at the APM / FPA planes as resolved in Figs. 4d and 4e, respectively. Despite the increased distance a clear contrast in APM power as well as SPAD counts is visible. Following the alignment of MU and OHS, the DL G.hn connection has been successfully established at a line rate of 833 Mb/s. We confirmed a stable connection over a duration of more than 35 minutes, with an average TPC throughput of 603 Mb/s (Fig. 4f). The few drops in data rate are a result of beam blocking as taller vehicles pass by and obstruct the direct line-of-sight between MU and OHS. Nevertheless, the otherwise constant throughput confirms reliable operation over the OWC channel.

Conclusions

We have demonstrated an OHS for wavelength-tuned beamsteering and beam spot adaptation to integrate user localization, as shown over a 107-m out-door OWC link. Further optimization of the MEMS mirror and FPA lens sizes are expected to further enhance the optical link budget (or OWC rate) and the field-of-view of the OHS. Real-time MU tracking is left for future work.

Acknowledgements

This work has received funding from the Smart Networks and Services Joint Undertaking (SNS JU) under the European Union's Horizon-Europe research and innovation programme under Grant Agreement No. 101139182.

References

- [1] C.W. Chow, "Optical Wireless Communication – Recent Progresses and Future Perspectives," *Journal of Lightwave Technology*, vol. 44, no. 3, pp. 806-820, 2026. DOI: 10.1109/JLT.2025.3607690
- [2] Y. Wang, G. Zhou, X. Zhang, K. Kwon, P.A. Blanche, N. Triesault, K.S. Yu, and M.C. Wu, "2D broadband beamsteering with large-scale MEMS optical phased array," *Optica*, vol. 6, no. 5, pp. 557-562, 2019. DOI: 10.1364/OPTICA.6.000557
- [3] Y. Liu, and H. Hu, "Silicon optical phased array with a 180-degree field of view for 2D optical beam steering," *Optica*, vol. 9, no. 8, pp. 903-907, 2022. DOI: 10.1364/OPTICA.458642
- [4] W. Ren, P. Hou, Y. Jiang, X. Zeng, and J. Sun, "100 km One-Point to Multi-Point Simultaneous Laser Communication Based on 8×8 Matrix Vector Optical Phased Array," *Journal of Lightwave Technology*, vol. 43, no. 17, pp. 8029-8039, 2025. DOI: 10.1109/JLT.2025.3581583
- [5] Y.Z. Lin, C.W. Chow, Y.C. Chang, C.Y. Hsu, T.C. Wei, P.Y. Hsieh, H.S. Chiu, S.Y. Lee, P.C. Kuo, K.J. Yang, Y.H. Hong, and H.C. Kuo, "2D Silicon Photonic Metalens-Based Focal Plane Array (FPA) Optical Beam Steering With Wide Field-of-View (FOV) for Chip-to-Chip Optical Wireless Communication (OWC)," *Journal Selected Topics in Quantum Electronics*, vol. 32, no. 2, art. no. 6100509, 2025. DOI: 10.1109/JSTQE.2025.3615889
- [6] X. Ma, M. Yuan, J. Li, H. Zhang, B. He, P. Zhang, Y. Jiang, H. Xiao, G. Ren, A. Mitchell, Y. Su, and Y. Tian, "Multi-target and ultra-high-speed optical wireless communication using a thin-film lithium niobate optical phased array," *Nature Communications*, vol. 17, art. no. 969, 2026. DOI: 10.1038/s41467-025-67696-3
- [7] Z. Yuan, J. Chen, Y. Wang, Y. Wu, M. Li, S. Chen, C. Chen, X. You, and Z. Zhang, "Shape-optimized metasurface beamformer for high-efficiency full-duplex optical wireless communications across an ultra-wide field-of-view," *Nature Communications*, early access, 2026. DOI: 10.1038/s41467-026-70665-z
- [8] P.V. Trinh, A. Carrasco-Casado, T. Okura, H. Tsuji, D.R. Kolev, K. Shiratama, Y. Munemasa, and M. Toyoshima, "Experimental channel statistics of drone-to-ground retro-reflected FSO links with fine-tracking systems," *IEEE Access*, vol. 9, pp. 137148-137164, 2021. DOI: 10.1109/ACCESS.2021.3117266.
- [9] Z. Li, Y. Li, Z. Zang, Y. Han, L. Wu, M. Li, Q. Li, and H.Y. Fu, "LiDAR integrated IR OWC system with the abilities of user localization and high-speed data transmission," *Optics Express*, vol. 30, no. 12, pp. 20796-20808, 2022. DOI: 10.1364/OE.454266
- [10] T. Koonen, K.A. Mekonnen, F. Huijskens, and E. Tangdiongga, "Bi-Directional All-Optical Wireless Gigabit Ethernet Communication System Using Automatic Self-Aligned Beam Steering," *Journal of Lightwave Technology*, vol. 41, no. 11, pp. 3446-3454, 2022. DOI: 10.1109/JLT.2022.3231438
- [11] S.M. Walsh, S. Karpathakis, A.S. McCann, B.P. Dix-Matthews, A.M. Frost, D.R. Gozzard, C.T. Gravestock, and S.W. Schediwy, "Demonstration of 100 Gbps coherent free-space optical communications at LEO tracking rates," *Scientific Reports*, vol. 12, art. no. 18345, 2022. DOI: 10.1038/s41598-022-22027-0
- [12] A. Willitsford, K.T. Newell, M. O'Toole, and K. Patel, "Maritime free space optical communications field test and link budget statistics," *Optics Express*, vol. 32, no. 8, pp. 13769-13782, 2024. DOI: 10.1364/OE.518363
- [13] F. Honz, and B. Schrenk, "Alignment-Tolerant Optical Fi-Wi-Fi Bridge Assisted by a Focal Plane Array Beamformer as Air Interface," *Journal of Lightwave Technology*, vol. 43, no. 13, pp. 6187-6193, 2025. DOI: 10.1109/JLT.2025.3553914
- [14] W. Liu, S. Chen, N. Huang, C. Gong, Y. Chen, and Z. Xu, "A Real-Time MEMS Mirror Based Beam Steering and Tracking System for Optical Wireless Communication," *Journal of Lightwave Technology*, vol. 43, no. 6, pp. 2681-2689, 2025. DOI: 10.1109/JLT.2024.3510452